

Zhenglin Li

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Education

University of Oxford

MEng Engineering Science

Oct 2022 – July 2026

Technical Skills

Optics & Photonics: Spatial Light Modulators (SLM), Polarisation Control, Waveguide Optics, Structured Light, Polarimetry, Optical System Design

Simulation & Modelling: MATLAB, Simulink, Coupled Mode Theory (CMT), Numerical Simulation, Eigenmode Analysis

Experimental: Optical Alignment, Fibre Coupling, Stokes Measurement, System Calibration, Femtosecond Laser Writing

Programming: Python, MATLAB, Java, PyTorch(Leaning)

Engineering Tools: SolidWorks, CAD

Research Projects

Master Project: Full Vectorial Control of Arbitrary Optical Fields

Oct 2025 – July 2026

Research Project(Supervised by Prof. Patrick Salter and Prof. Chao He)

University of Oxford

- Designed a multi-SLM optical system for full vectorial control of optical fields, enabling simultaneous modulation of intensity, phase, and polarisation
- Developed 4-SLM architecture and identified phase discontinuity limitations via Jones formalism and Poincaré sphere analysis
- Engineered a 5-SLM system introducing an additional degree of freedom to eliminate topological discontinuities
- Implemented simulation and machine learning optimisation pipeline for continuous phase pattern generation in feedback-driven systems
- Integrated optical system components (4f system, beam expansion, polarisation measurement) for experimental validation
- Tech:** SLM, Optical Systems, Polarisation Control, Machine Learning, Waveguide Optics

Vectorial Optics and Photonics Group

Jun 2025 – Aug 2025

Summer Research Intern(Supervised by Prof. Chao He)

University of Oxford

- Self-Project: Optical Skyrmion in Dielectric Waveguides**
- Built a complete optical system for structured light generation, including vector beam generation, fibre coupling, and polarisation-resolved measurement
- Derived eigenmodes of dielectric waveguides and implemented numerical simulations for modal analysis
- Applied topological analysis (Skyrmion representation) to characterise polarisation structures and compute invariants
- Explored topological mode encoding in single- and multi-mode fibres for high-dimensional optical communication
- Validated simulations with experimental measurements, demonstrating topology-based mode robustness
- Tech:** Waveguide Optics, Structured Light, Fibre Optics, Numerical Simulation, Topological Photonics

Dynamic Optics and Photonics Group

Jun 2024 – Aug 2024

Undergraduate Research Opportunities Programme(Supervised by Prof. Martin J. Booth and Prof. Patrick Salter)

University of Oxford

- Self-Project: Investigating the Use of Couplers for Polarimeter Application**
- Designed and fabricated 40+ femtosecond laser-written directional couplers for polarimetry applications
- Developed CMT-based models and validated with 500+ measurements, achieving 1.3% MSE
- Analysed polarisation-dependent coupling and proposed methods for polarisation-independent operation
- Co-authored paper in *Optics Communications*
- Tech:** Directional Couplers, CMT, Polarimetry, Laser Fabrication

Experience

Industrial Visiting

Aug 2022

Vital Thin Film Materials Co. Ltd.

Guangzhou, China

- Establish connections with enterprise executives and technical staff
- Gained firsthand exposure to semiconductor manufacturing processes through detailed observation of production lines.
- Studied product specifications related to sputtering targets and deposition materials.
- Exposed to industrial supply chains and downstream market integration.
- Tech:** Semiconductor Manufacturing, Thin Films, Materials Processing, Supply Chain Analysis

Software Engineer Intern

Aug 2019

Tencent

Shenzhen, China

- Collaborated with software engineers on code development and algorithm design tasks.
- Strengthened GitHub-based version control and collaborative development workflows.
- Tech:** Software Development, Algorithms, GitHub, Team Collaboration

Publications

ZhiKai Pong, Bangshan Sun, **Zhenglin Li**, et al.

Polarization-based modulation of splitting ratio in femtosecond laser directional couplers.

Optics Communications

An Aloysius Wang, Yuxi Cai, **Zhenglin Li**, et al.

General in situ feedback control of cascaded liquid crystal spatial light modulators for structured field generation.

arXiv:2601.11405 (under review at Laser & Photonics Reviews)

Yuxi Cai, Tingxian Gao, **Zhenglin Li**, et al.

Rethinking conditioning in polarimetry: a new framework beyond ℓ^2 -optimality.

arXiv:2512.23689(under submission)

Ruofu Liu, An Aloysius Wang, Yunqi Zhang, Yuxi Cai, Yihan Liu, **Zhenglin Li**, et al.

Topologically robust programmable logic arrays using light and matter skyrmions.

arXiv:2602.01455(Nature Communication)

An Aloysius Wang, **Zhenglin Li**, Yuxi Cai, Yifei Ma, Chao He, Bowei Dong.

Robust skyrmion transport in waveguides via topologically stabilizing modes.

Under review at *Physical Review A*

Honors

EUROP Research Prize

Department of Engineering Science, University of Oxford

- Selected as one of 10 students across the department for a competitive research-based programme
- Awarded for outstanding academic performance and research potential in engineering science

Undergraduation Scholarship Award

Mansfield College, University of Oxford

- Awarded for consistently excellent academic performance and strong scholarly potential(rank top 30).
- Includes a financial award of £200 per annum.

Extracurricular Leadership

Team Leader, Advanced Photonics Platform Design Project

Oct 2024 – Jun 2025

University of Oxford

- Led a team of 6 engineers to design a photonics platform for super-resolution imaging and laser fabrication
- Coordinated system-level design across optical modulation, scanning control, and mechanical integration
- Directed development of control systems using Simulink for synchronised optical and scanning processes
- Managed project planning, task allocation, and technical integration across multidisciplinary components
- **Tech:** Leadership, System Design, Simulink, Optical Engineering, Project Management